

Spread-Out Bragg Peak (SOBP) field design for the Stage-A proton-transport simulation

Method note for review

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Abstract

This note documents the method used to build a Spread-Out Bragg Peak (SOBP) in the Stage-A simulation, before it is implemented. A single mono-energetic pencil deposits a sharp Bragg peak and paints a thin track, giving an unrealistically small integral activity. A clinical field instead spreads dose uniformly over a target volume, which is what fixes the realistic absolute β^+ yield (and lets us cross-check against Parodi *et al.* 2008, Table 2). We construct the depth component of such a field analytically: a weighted superposition of pristine Bragg peaks whose weights flatten the dose over the target’s depth extent. The energies and weights are computed in Python (`field_design/sobp.py`) and written to a layer table that the Geant4 gun samples from; the resulting plateau is then verified directly against the simulated depth–dose curve. Lateral spreading and dose normalisation are separate steps, noted at the end.

1 Goal and scope

We want a proton field that deposits a **flat dose plateau** across the target’s depth extent $[R_p, R_d]$ (proximal to distal range), so that a prescribed dose can be painted *uniformly* over the target. This note covers only the **depth** component (the SOBP proper). The target geometry, its size and the prescribed dose are all **parameters** (Sec. 5); nothing here is hard-wired to the Parodi reference numbers.

2 Pristine Bragg peak and the range–energy relation

A mono-energetic proton of energy E stops at a well-defined depth, its **range** R . To good approximation the two are related by the Bortfeld power law [1],

$$R(E) = \alpha E^p, \tag{1}$$

with, in water, $\alpha = 2.2 \times 10^{-3} \text{ cm MeV}^{-p}$ and $p = 1.77$. For another homogeneous material we scale the range by the (mass) stopping-power ratio, which to first order is the inverse mass-density ratio, $R_{\text{mat}} \simeq R_{\text{water}}/\rho_{\text{rel}}$ ($\rho_{\text{rel}} \approx 1.04$ for brain). The energy needed to reach a depth R is the inverse of Eq. (1), $E(R) = (R/\alpha)^{1/p}$.

Equation (1) is used only to choose the *layer energies*; the actual peak shapes come from Geant4, and the plateau is verified there (Sec. 7), so small inaccuracies in α, p are corrected by the verification step.

A useful consequence of Eq. (1): a proton that has travelled to depth $z < R$ has residual range $R - z$ and residual energy $\propto (R - z)^{1/p}$, so its mass stopping power (dose per unit length) scales as

$$S(z; R) \propto \frac{dE}{dz} \propto (R - z)^{1/p-1}. \tag{2}$$

This is the only feature of the peak shape the weight derivation needs.

3 SOBP as a weighted superposition

We place $N+1$ pristine peaks at ranges R_i spanning the target depth,

$$R_i = R_p + \frac{i}{N} (R_d - R_p), \quad i = 0, \dots, N, \quad (3)$$

($R_0 = R_p$ proximal, $R_N = R_d$ distal), with relative fluence weights $w_i \geq 0$. The total depth-dose is their superposition,

$$D_{\text{SOBP}}(z) = \sum_{i: R_i \geq z} w_i S(z; R_i). \quad (4)$$

The gun realises Eq. (4) stochastically: each primary proton is assigned to layer i with probability $\propto w_i$ and given energy $E_i = E(R_i)$.

4 Analytic flattening weights

We want $D_{\text{SOBP}}(z) = \text{const}$ for $z \in [R_p, R_d]$. Treating the range as continuous with fluence density $\Phi(R)$, Eq. (4) with (2) becomes the Abel-type integral

$$D(z) = \int_z^{R_d} \Phi(R) (R - z)^{1/p-1} dR \stackrel{!}{=} \text{const}. \quad (5)$$

This is an Abel (right-sided fractional) integral equation of order $a = 1/p$; inverting it for a constant left-hand side gives

$$\boxed{\Phi(R) \propto (R_d - R)^{-1/p}} \quad R \in [R_p, R_d], \quad (6)$$

plus a finite reinforcement at the proximal edge $R = R_p$ that fills the proximal corner of the plateau. Since $-1/p \approx -0.565 < 0$, the weight rises towards the *distal* peak (which no deeper peak reinforces) and diverges integrably at $R = R_d$. Integrating Eq. (6) over each range bin $[R_i - \frac{\Delta}{2}, R_i + \frac{\Delta}{2}]$ (with $\Delta = (R_d - R_p)/N$) gives the finite discrete weights actually used,

$$w_i \propto (R_d - R_i + \frac{\Delta}{2})^{1-1/p} - (R_d - R_i - \frac{\Delta}{2})^{1-1/p}, \quad (7)$$

with the distal bin truncated at R_d and an added proximal term for $i = 0$. The weights are normalised to sum to one and stored as a probability table.

Why analytic (vs. fit-to-simulation). Equation (7) needs no simulation and is the standard, well-validated construction. We adopt it as the baseline; if the verified plateau (Sec. 7) is not flat enough, the fallback is to replace the analytic $S(z; R_i)$ with Geant4-simulated pristine peaks and solve Eq. (4) for w_i by non-negative least squares.

5 Parameters

Everything that defines the target and the field is a parameter:

symbol	meaning	Parodi-standard value
$d_{\text{prox}}, d_{\text{dist}}$	target proximal/distal depth	$\approx 5.5, 10.5$ cm
$\varnothing_t$	target (tumour) diameter	6 cm
$L_t = d_{\text{dist}} - d_{\text{prox}}$	target length (modulation width)	5 cm
D	prescribed dose to the target	1 Gy
$N+1$	number of energy layers	$\sim 15\text{--}25$
material	phantom material	G4_BRAIN_ICRP

The depths map to ranges R_p, R_d in the chosen material via Sec. 2.

6 Algorithm and output

1. From $(d_{\text{prox}}, d_{\text{dist}}, N, \text{material})$ compute R_p, R_d and the layer ranges R_i .
2. Energies $E_i = E(R_i)$ from Eq. (1); weights w_i from Eq. (7); normalise $\sum_i w_i = 1$.
3. Write `data/sobp_layers.csv` with columns `energy_MeV`, `weight`, and a sanity plot of the analytic SOBP.
4. The Geant4 gun loads the table and samples one layer per primary.

7 Verification

We run Stage A with the SOBP gun and inspect the simulated depth–dose (`depth_dose.csv` via `bragg_profile.py`). Acceptance: the dose is flat to within a few percent across $[R_p, R_d]$ with a sharp distal fall-off at R_d . If not, increase N or fall back to the least-squares weights above.

8 Out of scope here (next steps)

- **Lateral spreading:** uniform fluence over the target cross-section (diameter $\varnothing_t$), turning the depth plateau into a dose box.
- **Dose normalisation:** score dose in the target volume and scale the number of primaries so the target dose equals D ; this fixes the absolute P_j .
- **Cross-check:** compare the integral β^+ yields to Parodi Table 2 ($\sim 1.8 \times 10^8$ decays per Gy, head).

References

- [1] T. Bortfeld and W. Schlegel, *An analytical approximation of depth–dose distributions for therapeutic proton beams*, Phys. Med. Biol. **41**, 1331 (1996); see also T. Bortfeld, Med. Phys. **24**, 2024 (1997).